Imperial College London

Department of Earth Science and Engineering

MSc in Geo-Energy with Machine Learning and Data Science

Independent Research Project

Project Plan

Developing Machine Learning Models Using Crystal Textures to Investigate Magmatic Evolution Leading to Explosive Volcanic Eruptions

by

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Abstract

1. Introduction

1.1 – Introduction/Problem Outline

1.2 - Methodology

1.3 - Literature Review

Literature Review Subheading 1 (Minor Topic: The Geology/Microscopy Related to Image Data)

Literature Review Subheading 2 (Major Topic: Improving Segmentation)

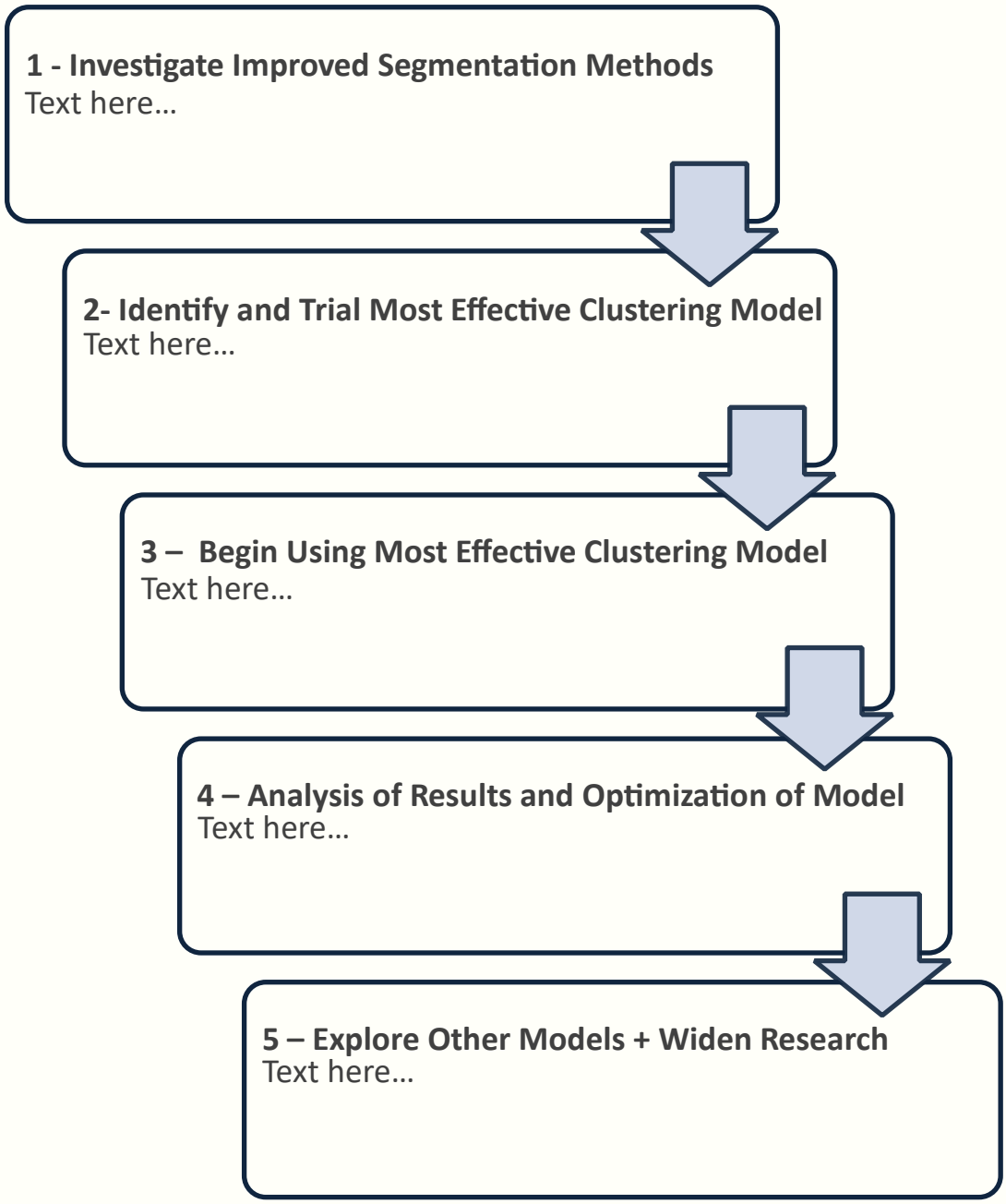
Literature Review Subheading 3 (Major Topic: Unsupervised Clustering)

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2. Progress at the Time of This Report

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3. Plan for the Rest of This Project



(Above) Flow chart outlining major steps left for the research project.

References

30 References in the style given below… [8/30 so far…] ALPHABETICAL

Haller, L.L. (n.d.). *Investigating magmatic processes using cold-cathodoluminescence imaging of plagioclase*. [online] Available at: https://tinyurl.com/4s8erysv [Accessed 26 May 2026].

(Haller, n.d.)

Halverson, B., Loewen, M., Dietterich, H. and Whittington, A. (2025). Case study of deep learning image segmentation for the purposes of rapid 2D petrographic analysis in volcanic rocks. *Volcanica*, [online] 8(2), pp.427–442. doi:https://doi.org/10.30909/vol/gsfc1696.

(Halverson et al., 2025)

Hassan, S.M., Laban, N., Abo Khashaba, S.M., El-Shibiny, N.H., Bashir, B., Azer, M.K., Drüppel, K. and Keshk, H.M. (2024). Semantic Segmentation of Some Rock-Forming Mineral Thin Sections Using Deep Learning Algorithms: A Case Study from the Nikeiba Area, South Eastern Desert, Egypt. *Remote Sensing*, [online] 16(13), p.2276. doi:https://doi.org/10.3390/rs16132276.

(Hassan et al., 2024)

Jutzeler, M., Carey, R.J., Dagasan, Y., McNeill, A. and Cas, R.A.F. (2024). Machine-learning crystal size distribution for volcanic stratigraphy correlation. *Scientific Reports*, [online] 14(1). doi:https://doi.org/10.1038/s41598-024-82847-0.

(Jutzeler et al., 2024)

Koeshidayatullah, A. and Ferreira, I. (2023). SelfMin: Self-Supervised Deep Learning for Advanced Mineralogical Analysis. *EGU General Assembly 2023*. [online] doi:https://doi.org/10.5194/egusphere-egu23-2189.

(Koeshidayatullah and Ferreira, 2023)

Leichter, A., Almeev, R.R., Wittich, D., Beckmann, P., Franz Rottensteiner, Holtz, F. and Sester, M. (2022). Automated Segmentation of Olivine Phenocrysts in a Volcanic Rock Thin Section Using a Fully Convolutional Neural Network. *Frontiers in Earth Science*, [online] 10. doi:https://doi.org/10.3389/feart.2022.740638.

(Leichter et al., 2022)

Morales, J., Saldivia, C., Lobo, R., del Pino, M., Muñoz, M., Caniggia, G., Catalán, J., Hayde, R. and Poblete, V. (2025). Visual analysis of deep learning semantic segmentation applied to petrographic thin sections. *Scientific Reports*, [online] 15(1). doi:https://doi.org/10.1038/s41598-025-99767-2.

(Morales et al., 2025)

Toth, N. and Maclennan, J. (2024). MinDet1: A deep learning-enabled approach for plagioclase textural studies. *Volcanica*, [online] 7(1), pp.135–151. doi:https://doi.org/10.30909/vol.07.01.135151.

(Toth and Maclennan, 2024)

AI Acknowledgement

Service - Chat-GPT, Claude

Producer - Open AI, Anthropic

URL - https://chatgpt.com, https://claude.ai

How generative AI was used - Write Just Before Submission

Declaration regarding AI use - All work in this project plan has been produced and written solely by the author listed above despite the use of the Generative AI tool listed above.

WORKING NOTES - DELETE BEFORE SUBMISSION

Notes Page – Delete Before Submission

Document Formatting:

#404040 – Text Colour Code

Project Plan Limits:

Word Limit: 1,500 Words Max.

Figure + Table Limit: 5 max.

Figures to be included:

1. *Progress Flowchart*
2. *Crystal Segmentation Example??*

3. *At Least one for Methodology*
4. *Model Frameworks?/Pipelines*
5. *??*

Max. .PDF File Size: 4MB

Suggested Models to Look into (Self-Supervised Learning – SSL):

- SimSiam
- SimCLR
- DINOv3
- DeepCluster